

Problem Set 1

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Problem 1:

Prove that $4 \nmid (n^2 + 2)$ for any integer n .

Solution:

Let $n \in \mathbb{Z}$ be any arbitrary integer. To prove that $n^2 + 2$ isn't divisible by 4, (or notationally, that $4 \nmid (n^2 + 2)$), we'll proceed via a proof by contradiction.

So, for the purposes of deriving a contradiction, let's assume that there just so happened to exist some integer, $m \in \mathbb{Z}$, such that $m^2 + 2$ was divisible by 4. Then, by definition of divisibility, we know that there must exist an integer, $k_1 \in \mathbb{Z}$ for which $4k_1 = m^2 + 2$. Notice, however, that this must mean $m^2 + 2$ is an even number, since 2 is a divisor of 4 and, as we just assumed, $m^2 + 2$ is itself divisible by 4. Therefore we know once again, by definition of divisibility, that there must exist some other integer k_2 satisfying $2k_2 = m^2 + 2$. By performing some very simple manipulations, we may observe that:

$$\begin{aligned} 2k_2 &= m^2 + 2 && \text{(By our earlier equality)} \\ \implies 2k_2 - 2 &= m^2 && \text{(Subtracting 2 from both sides)} \\ \implies 2(k_2 - 1) &= m^2 && \text{(By distributivity of multiplication over addition)} \end{aligned}$$

Which clearly means that m^2 is an even number as well! It is a well established fact that a square being an even integer implies that its square root is even too, but to quickly prove it (for thoroughness' sake, and so I don't get docked marks!), consider some arbitrary integer $a \in \mathbb{Z}$ and assume that its square, a^2 , is even. Invoking the definition of divisibility again, we therefore know that there exists some constant $l_1 \in \mathbb{Z}$ for which $a^2 = 2l_1$. We thus have that:

$$\begin{aligned} a^2 &= 2l_1 && \text{(By our earlier equality)} \\ \implies a &= \pm\sqrt{2l_1} && \text{(Applying the square root to both sides)} \\ &= \pm\sqrt{2}\sqrt{l_1} && \text{(By distributivity of exponents over multiplication)} \end{aligned}$$

But remember! a was assumed to be an integer! This naturally must mean that $\sqrt{2}\sqrt{l_1} \in \mathbb{Z}$, which would only make sense if l_1 could be expressed as a multiple of 2 (since otherwise, a would be irrational, and therefore not an integer). Hence, there exists some $l_2 \in \mathbb{Z}$ satisfying $l_1 = 2l_2$, meaning that:

$$\begin{aligned} a &= \pm\sqrt{2}\sqrt{l_1} && \text{(By our earlier equality)} \\ &= \pm\sqrt{2}\sqrt{2l_2} && \text{(By our characterization of } l_1) \\ &= \pm\sqrt{2}\sqrt{2}\sqrt{l_2} && \text{(By distributivity of exponents over multiplication)} \\ &= \pm 2\sqrt{l_2} && \text{(Simplifying)} \end{aligned}$$

And since a is an integer by hypothesis, we have that $\sqrt{l_2}$ must be an integer, implying that $a = \pm 2l_3$ for some integer l_3 , making a even by definition of evenness and thereby proving my prior claim that if m^2 is even, then so is m .

Continuing on with our original proof, now that we know that m^2 being even implies that m is even as well, we can (for the last time) assert that there exists some integer, $k_3 \in \mathbb{Z}$ such that $2k_3 = m$. It is here that we can now, finally, derive our contradiction. Notice that, using this characterization of m , we can go back to our original equation and deduce that:

$$\begin{array}{ll} 4k_1 = m^2 + 2 & \text{(By our choice of } k_1\text{)} \\ \implies 4k_1 = (2k_3)^2 + 2 & \text{(By our characterization of } m\text{)} \\ \implies 4k_1 = 4k_3^2 + 2 & \text{(Expanding)} \\ \implies 2k_1 = 2k_3^2 + 1 & \text{(Dividing both sides by 2)} \end{array}$$

Which is patently ridiculous! “Why?” I hear you ask? Well, because in that final equality, the left-hand side is clearly an even quantity, whereas the right-hand side is clearly an odd one. This makes no sense, as a number cannot be both even and odd! Hence, our assumption that some integer m existed such that $m^2 + 2$ was even must’ve been wrong.

Therefore, since no such m can possibly exist, we therefore have that $4 \nmid (n^2 + 2)$ for all possible $n \in \mathbb{Z}$.